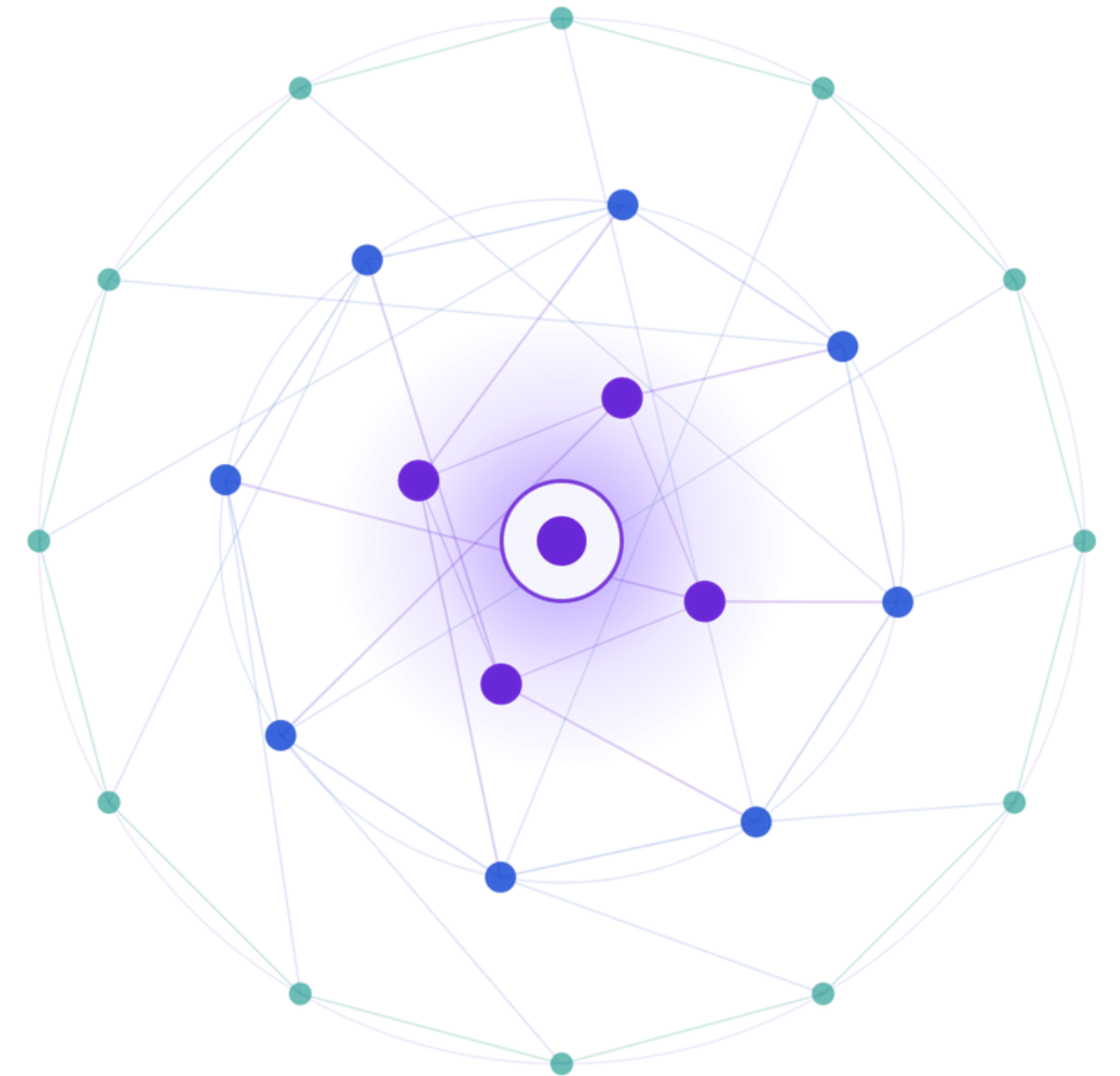


SAMSUNG PRISM

Generative AI Hackathon

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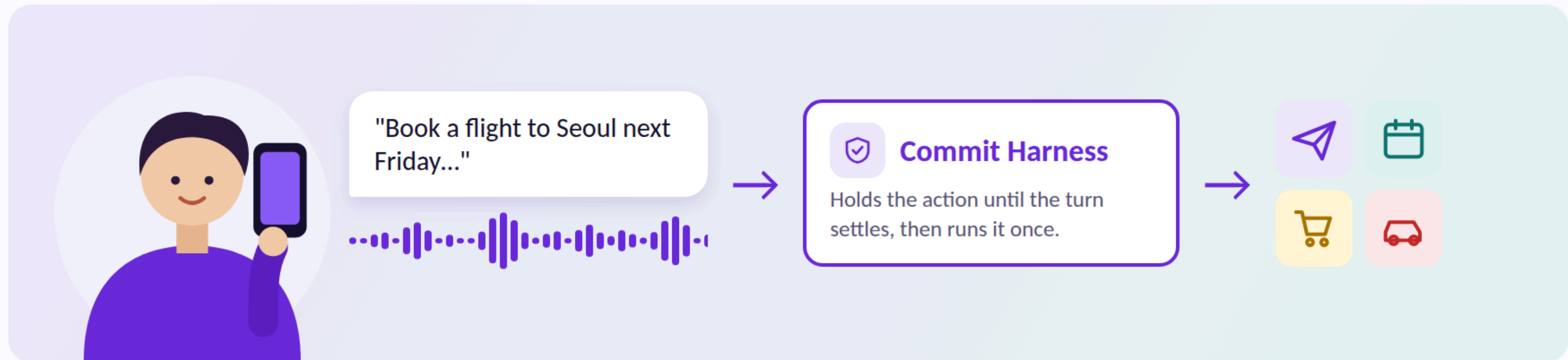
- Theme ID - 5 (INTERRUPTIBLE AGENTS)
- Team Name - **REIGN**
- College Name – **SRM INSTITUTE OF SCIENCE & TECHNOLOGY**
- Member Name & Email 1- **POKALA SAI NITHIN** sp9034@srmist.edu.in
- Member Name & Email 2- **ARYAN GARG** ag2841@srmist.edu.in
- Member Name & Email 3- **LOHITASHWA** ls0885@srmist.edu.in
- Member Name & Email 4- **V PREETHA** pv7486@srmist.edu.in
- Submission Github link - <https://github.com/Nithyon/Theme5-Interruptible-Agents>



THEME : 5 INTERRUPTIBLE AGENTS

Reliable and efficient **real-time voice agents**

Act too early and the agent books the wrong thing. Wait too long and it feels slow.



Act fast vs act safe

Two competing requirements. We make the trade-off explicit and measure its cost.



Propose → Settle → Commit

The action is held, not run. A quick check judges if the user is done.



Recover after execution

Retry, de-duplication, undo and human hand-off for late corrections.

62 → 67

judged pass out of 100, stock vs ours (Full-Duplex-Bench v3)

3.9 s → 5.3 s

reply delay, the price of waiting before actions

Existing Solutions & Gaps

Prior work improves turn-taking and repair, but most methods optimize conversational timing rather than a safe execution boundary for external actions.

1

Fixed Silence / VAD Thresholds

- Respond after a predefined silence period.
- Common systems wait approximately 700–1000 ms.
- Simple, inexpensive, and easy to deploy.



!

Gap: A fixed timer cannot distinguish hesitation from the true end of a request. Short thresholds cause premature actions; long thresholds slow every interaction.

Li et al., SIGDIAL 2022.

2

Acoustic and Prosodic Turn Prediction

- Uses voice activity, pitch, rhythm, and speaker interaction patterns.
- Voice Activity Projection predicts future speech activity continuously.
- More responsive than silence-only endpointing.



!

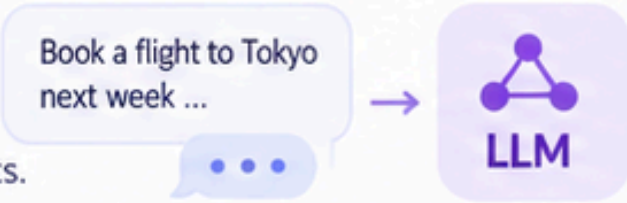
Gap: Primarily predicts who will speak next, not whether it is safe to execute a real-world action; performance can vary across languages and domains.

Ekstedt & Skantze, 2022.

3

Linguistic End-of-Turn Prediction

- TurnGPT uses dialogue context and linguistic completeness.
- Incremental language models project possible completion points.



!

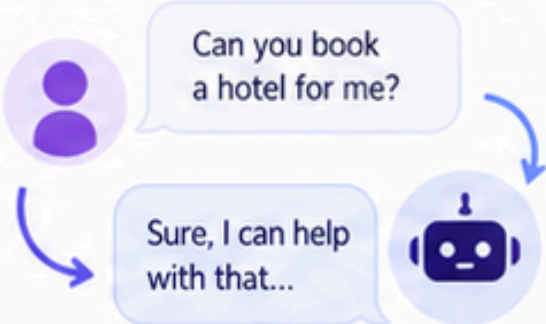
Gap: Depends on partial ASR transcripts and focuses on response timing; it does not define a transaction boundary before irreversible tool calls.

Ekstedt & Skantze, 2020; 2021.

4

Response-Conditioned Turn Taking

- Uses conversation history plus the response the agent intends to produce.
- Improves prediction when turn boundaries are ambiguous.



!

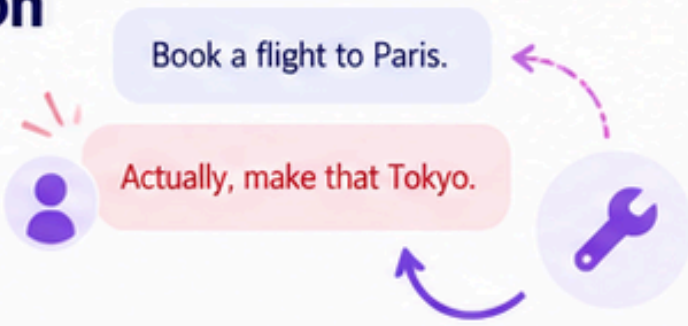
Gap: The objective is when the system should respond — not whether a booking, payment, message, or vehicle action should be committed.

Jiang et al., ACL Findings 2023.

5

Dialogue Repair and Correction

- Allows the agent to react after a misunderstanding or user correction.
- Repair datasets evaluate recovery from incorrect interpretations.



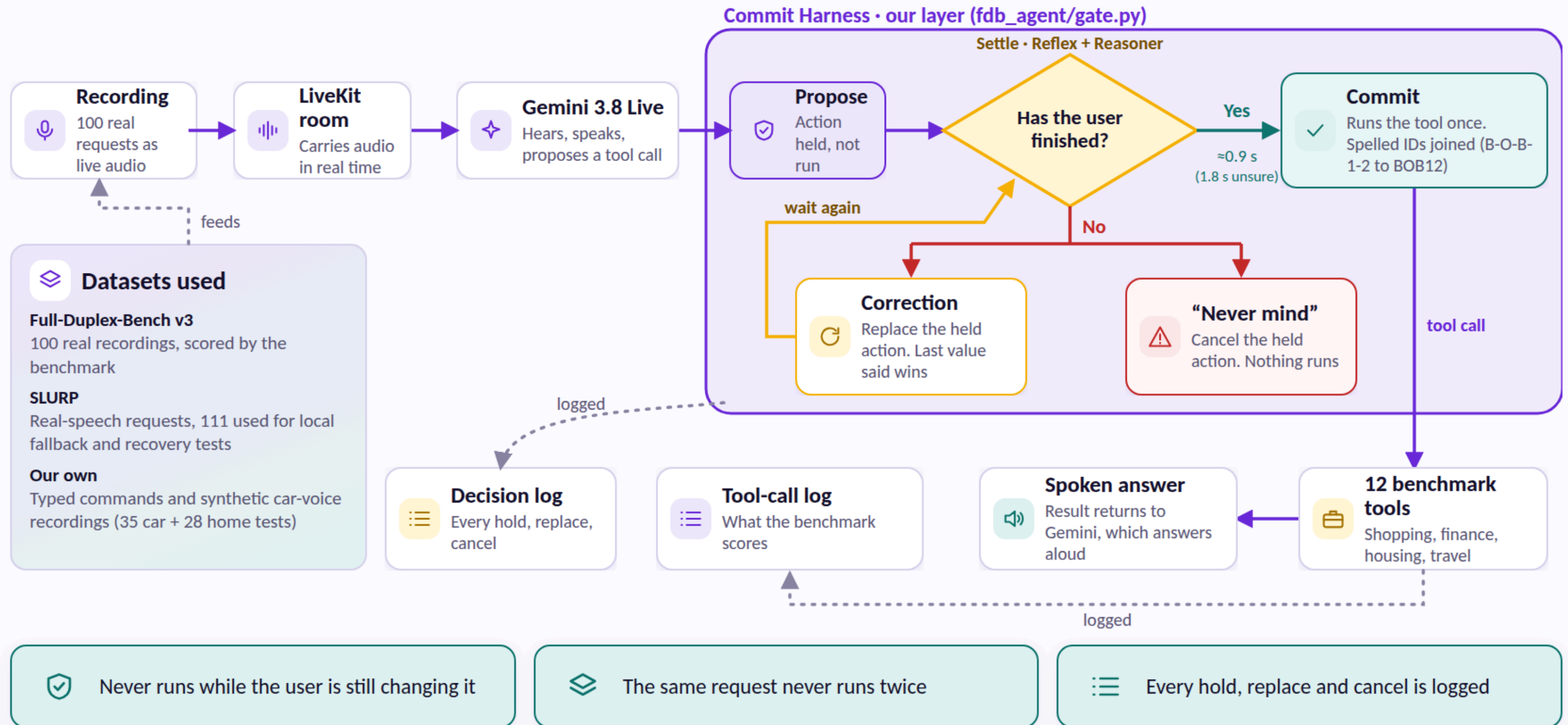
!

Gap: Models remain below humans on repair tasks, and dialogue repair does not itself provide retry, deduplication, undo, or rollback for actions that have already executed.

Chiyah-Garcia et al., EMNLP 2024

Our Solutions & Architecture Diagram

How one request flows through the submitted benchmark agent. The purple frame is the part we added.



DEMO

Show a premature-action failure, then the same request through the harness.



User speaks · "Book a flight to Boston..."



Model proposes · book_flight(Boston) held, not run



Harness settles · Reflex reads hesitation, Reasoner checks completeness



Commit · waits about 0.9 s (1.8 s if unsure). A correction replaces the held action



Tool runs once · only after the commit boundary



"Book a flight to Boston..." (*pause*) "... actually, make that New York."

Without the harness

Boston

BOS

book_flight ran on the first phrase

EXECUTED

wrong
flight

With the harness

Boston · replaced

~~BOS~~

held

New York

JFK

runs
once

Propose ≠ Execute



Decision log

Proposal, replacement, commit time



Tool log

Proof of what actually ran



Fallback

Behavior if the Reasoner is unavailable

Inside the harness: where JEV fits

JEV (TypeSafe JEV) is the Reasoner: one part of the Settle step, not the whole system. Smart Turn was tested and left off.

Part	What it does	File	In run
Prompt rules	Last value said wins, use values as spoken, never say "done" before it ran	gate_agent.py	On
Reflex	Reads "um" or "no, sorry". Waits 0.9 s, or 1.8 s if unsure	gate.py	On
Reasoner (TypeSafe JEV)	Small classifier: has the user finished? Past 0.8 s, Reflex decides alone	jev.py	On
Identifier rule	Joins "B-O-B-1-2" into "BOB12" before the tool runs	gate.py	On
Listener (Smart Turn v3.2)	Hears tone to tell a pause from an ending. Wrong too often	smart_turn.py	Off

One request through Settle: the held action waits out the window, and a correction replaces it.



Tools and tech stack used

Real-time Voice & Core Model



LiveKit

Real-time audio transport and interaction pipeline.



Gemini Live

Primary conversational voice model. Listens, responds and proposes tool calls.



Action Control (Commit Harness)



Commit Harness

Custom action-control layer implementing Propose → Settle → Commit.



Reflex Module

Lightweight hesitation / lexical cue detector (conservative fallback).



TypeSafe JEV

Semantic end-of-turn reasoner to decide when the user's turn is complete enough for earlier execution.



Local Fallback (On-Device)



Ollama

Local model runtime for fallback experiments.



Gemma 4 26B

Stronger local fallback model (reliable on real requests).



Gemma 4 e4b / 4B-class

Smaller local model (faster but less reliable on several real-request cases).



Whisper / faster-whisper

Local speech-recognition path for fallback.



Tool Execution (Simulated Environment)



Mock Tool Layer

Simulated travel, maps, shopping, EV/vehicle, smart-home and related tool actions for safe evaluation.



Post-Commit Recovery



Recovery Layer

Retry, deduplication, undo/rebook and human-handoff logic for post-commit failures or corrections.



Evaluation, Datasets & Infrastructure



Full-Duplex-Bench v3

Main real-time voice-agent benchmark for evaluation.



SLURP

Real-speech dataset used in local fallback and recovery experiments.



Decision Logs + Tool Logs

Per-recording observability for proposed actions, commit timing, execution and recovery behavior.

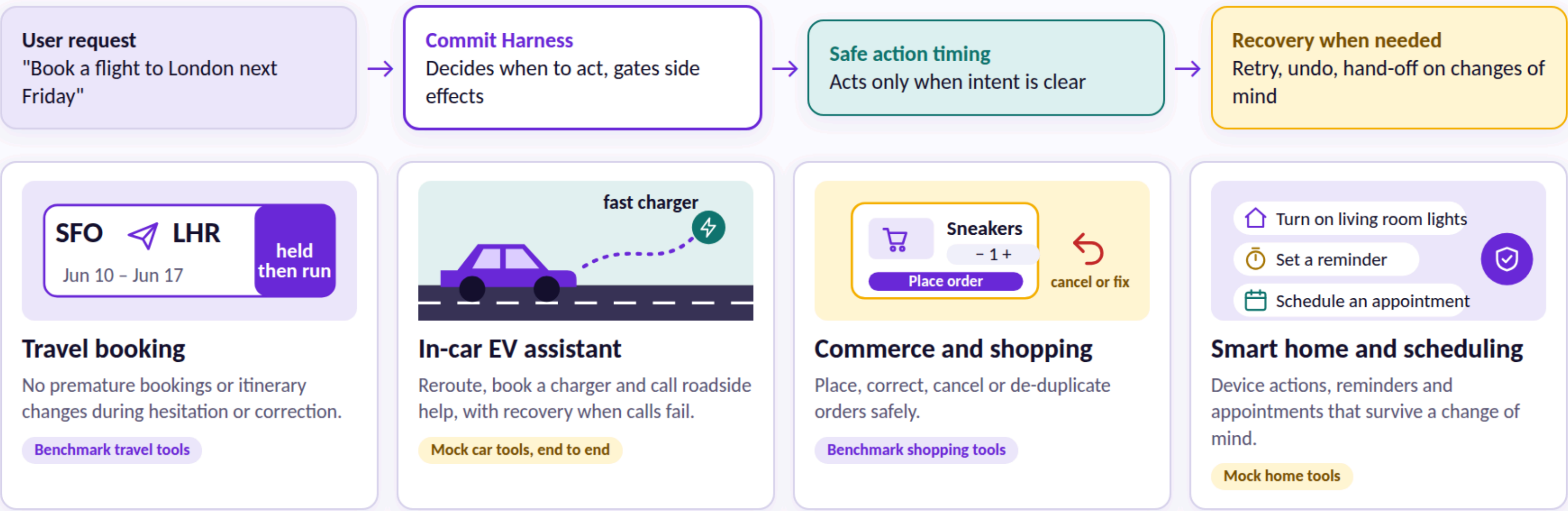


GitHub + reproduce.sh

Repository, experiment artifacts and reproducible setup instructions.

Impact & Use case

Any voice agent that can trigger a side effect needs to know when the user is done, and what to do when they change their mind.



67 judged pass out of 100 on Full-Duplex-Bench v3, up from 62

8/8 failed tool calls recovered in the real-speech run

63 offline tests pass: 35 car + 28 home

Innovation highlights, results and limitations

A commit harness with JEV for reliable, efficient and recoverable real-time voice agents.

Innovation Highlights

- 1

**Commit Harness with JEV**
Uses end-of-turn evidence (hesitation, prosody, semantic completion) to decide when to act. Executes earlier when confidence is high.
- 2

**Prevents premature actions**
Holds the proposed action during brief pauses and user edits. Reduces incorrect tool calls.
- 3

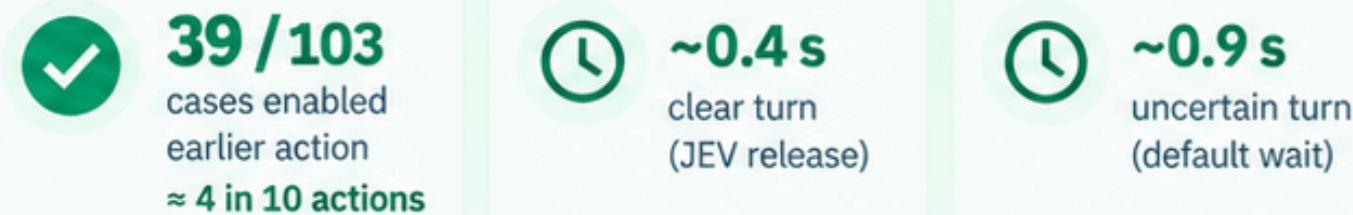
**Built-in recovery mechanisms**
Retry, deduplication, undo/revert and human handoff for corrections after execution.
- 4

**Hybrid local + cloud setup**
Works with cloud models (Gemini 3.8 Flash) and local fallback (Gemini 4 26B / 4 4B). Supports offline operation.
- 5

**Modular and extensible**
Domain-agnostic tool layer (travel, maps, shopping, smart home, etc.). Cost-aware model routing as future scope.

Results

Enables earlier and safer execution, with reliable recovery.



Evaluation on Interrupt Scenarios

Setup	Strict	Relaxed	Self-corrections	First tool call	Total
Local: Whisper → Gemini 4 26B	12/19	18/19	2/2	5.6 s	8.2 s
Local: Gemini 4 e4b (audio in)	10/19	13/19	2/2	4.0 s	5.5 s
Cloud: Gemini 3.8 Flash (audio in)	4/4 (of those that ran)	4/4	not reached	19–80 s	–

Detailed Interrupt Test Results

Model (Gemini 4 e4b)	Run	Right tool	Fully correct	Stayed out when no tool fits	Self-corrections	No answer	Median time
	1	29/29	29/29	6/8	14/16	0	2.24 s
	2	29/29	29/29	6/8	14/16	0	2.27 s
	3	29/29	29/29	6/8	14/16	0	2.12 s

Offline Fallback Suite (Gemini 4 26B)

Test	Right tool	Fully correct	Stayed out when no tool fits	Self-corrections	No answer	Median time
Own (typed)	36/36	34/36	4/4	6/6	0	5.56 s
SLURP (real)	46/51	45/51	60/60	0/0	0	5.99 s

Limitations

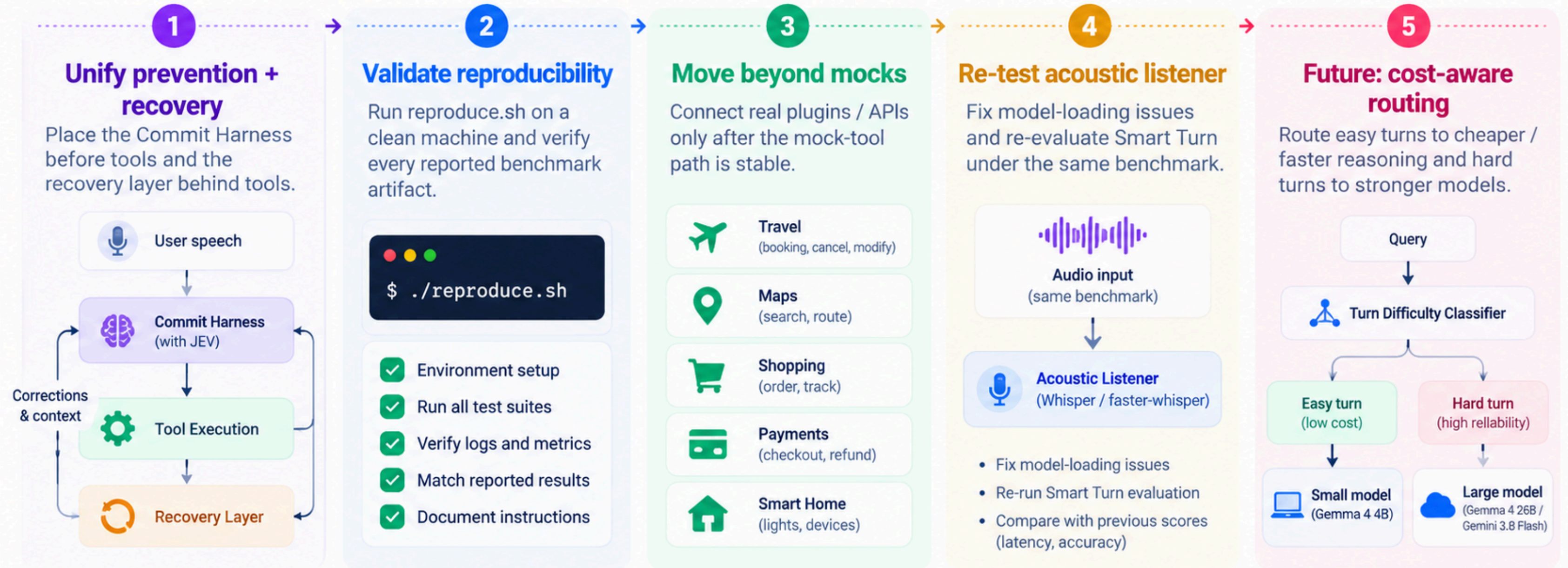
- **Prototype tools**
Uses mock tools. Not yet integrated with real services (e.g., Bixby, SmartThings).
- **Not a full agent yet**
Harness and recovery layer are not yet in a single end-to-end agent.
- **Model constraints**
Local fallback works reliably with large Gemini (4 26B) but smaller model (4 4B) fails on many real requests.
- **Evaluation scope**
Synthetic car voice and limited real recordings (111 SLURP). One run each for some setups. GPT-4o judge scoring not yet included.

Future Scope

- **Cost-aware model routing**
(small model for easy turns, larger model for hard turns).
- **Real plugins** for shopping, payments, maps, travel and SmartThings.
- **Stronger evaluation** with more real recordings and integration into a complete voice agent.

What's next

Turn the benchmark prototype into a complete prevention + recovery stack for real tool integrations.



Expected Outcome



Production-ready agent

Unified prevention + recovery with real tool integrations.



Reliable & reproducible

Consistent results across environments and datasets.



Real-world usefulness

Supports daily tasks (travel, payments, shopping, smart home).



Efficient & scalable

Cost-aware model routing and better user experience.

What's next: one agent that prevents and recovers

Turn the benchmark prototype into a complete prevention and recovery stack.



1 Unify both layers

Harness in front, recovery behind, so the benchmark agent can undo.



2 Ablate the gain

Test prompt rules and the identifier rule separately.



3 Cut the wait

Measure safe early release before 0.9 s.



4 Re-test the listener

Fix model loading and re-run Smart Turn.



5 Reproduce

Run reproduce.sh on a clean machine and match every number.



6 Beyond mocks

Real plugins and APIs. Cost-aware routing stays future scope.

Target architecture



User speech



Commit Harness



Tool execution



Recovery layer



late corrections go back to the harness

Brownie points slide(differentiation)

Recovery, Extension & Offline Fallback Evidence

Post-commit recovery handles late corrections, the EV extension shows product value, and local Gemma preserves reliability when the cloud is unavailable.

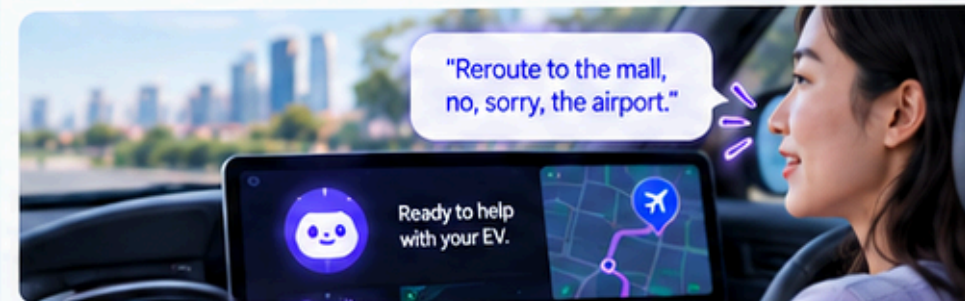
1 Recovery layer for post-commit changes

JEV prevents avoidable premature actions; recovery handles corrections that arrive after commitment.



2 Extension: an in-car EV assistant that recovers

A recovery layer handles reroutes, retries, deduplication, rebooking, and human handoff.



The driver says		What the recovery log shows	
1	 "Reroute to the mall, no, sorry, the airport."	→ 	one reroute, to the airport
2	 "Find a charging station near downtown."	→ 	two failures, quiet retries, found on the third try
3	 "Book that station for 6 pm." Then "Book it again."	→ 	one booking; the repeat is not re-run
4	 "Actually, make it 7 pm instead."	→ 	6 pm cancelled first, then 7 pm booked
5	 "Call roadside assistance." Twice; the line is down.	→ 	six attempts, then a hand-off to a human with a reference

Run end to end on audio

Recorded requests are played through LiveKit; replies and the log are saved.

 Also on real speech

11 real SLURP recordings,
lights tool failing first time:
10 of 11 carried out,
8 of 8 failures recovered.

Limits

Mock tools. The car voice is synthetic.
One run each.
35 + 28 offline tests pass.

3 When the cloud is gone: an offline Gemma fallback

Local fallback can preserve tool-use reliability when cloud models are unavailable.



Test (typed commands)	Gemma 4 26B 15.9 GB	Gemma 4 e4b 3.1 GB
Our 40 commands: right action and values	34/36	34/36
111 real requests (SLURP): light requests right	45/51	18/51
SLURP requests no tool fits: correctly did nothing	60/60	60/60
Self-interruptions: right action ("24, no, 21")	28/29	29/29
"Never mind": correctly did nothing	8/8	6/8
Time per command	about 6 s	about 2 s

>_ Local fallback suite (Gemma 4 26B)

Set	Right tool	Fully correct	Stayed out when no tool fits	Self- corrections	No answer	Median time
own	36/36	34/36	4/4	6/6	0	5.56 s
SLURP	46/51	45/51	60/60	0/0	0	5.99 s



The large model is safe to fall back on.

The small one is faster but refused many real requests and, after "leave them as they are", still turned the lights off.

Measured on an Intel Core Ultra 7 258V laptop with 32 GB RAM using Ollama.

 Example car tools



Reroute

Update the route to a new destination.



Charger search

Find nearby fast chargers
with real-time availability.



Charger booking

Reserve a charger at the selected time and location.



Roadside help

Request roadside assistance when needed.



Evidence:



35 car + 28 home
offline tests passed



In-car scenario
run end to end



Mock tools, not Bixby or SmartThings

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo **Yes**
- README with reproducible setup instructions **Yes**
- Demo video, max 5 minutes (YouTube or Drive link) **Yes**
- Presentation file (PPT or PDF) **Yes**

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India